



Indian Association for the Cultivation of Science
(Deemed to be University under *de novo* Category)
**Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's
Program/PhD Course**
Mid-Semester Examination-Autumn 2025

Subject: Fundamentals of molecular cell biology
Full Marks: 25

Subject Code(s): BIS 2101
Time Allotted: 2 h

Part-A

- 1. Define the following (any three):** **1×3=3**
 - a) Chromosome Territory
 - b) Virusoids
 - c) Muton
 - d) Plasmid

- 2. Explain (any one):** **2×1=2**
 - a) *Escherichia coli* chromosome has both positive and negative supercoiling in it.
 - b) Packaging of eukaryotic cell DNA is not static.

- 3. Answer the following (any two):** **3×2=6**
 - a) What is genome? How many types of DNA are present in the genome of a photosynthesizing plant cell? **1+2=3**
 - b) What is Linking number? One closed circular DNA has 10 twists while another closed circular DNA has 12 twists and 2 negative writhes. Find their difference in the degree of supercoiling. **2+1=3**
 - c) Describe beads on a string packaging of eukaryotic chromosomal DNA. What is the width of a chromosome during metaphase? **2+1=3**

- 4. Describe the following (any one):** **2×1=2**
 - a) Eukaryotic gene structure.
 - b) An experiment to identify the chemical nature of the bacterial genetic material

P.T.O

Part-B

Answer the following questions

6×2=12

1. Explain the process of capacitation in sperm and discuss its significance in successful fertilization.
2. Why is the technique of sperm washing considered crucial in assisted reproductive procedures such as IVF?
3. Describe the biological mechanisms that lead to the formation of identical and fraternal twins.
4. Compare and contrast mitosis and meiosis with respect to purpose, genetic outcome, and stages.
5. Justify the need for establishing a stem cell repository at the time of childbirth.
6. Illustrate the mammalian cell cycle, labeling the major phases and checkpoints involved.